CSM
Spring 2026

EECS 16B
Worksheet 9

1. Bode Plots for Filters

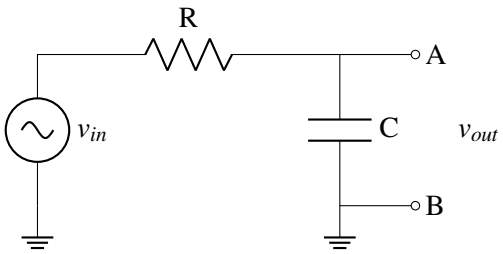


Figure 1: Low Pass Filter

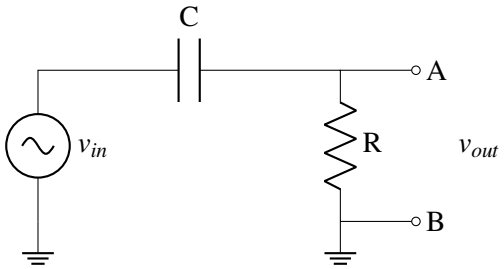


Figure 2: High Pass Filter

(a) First, consider the following transfer function of a Low-Pass Filter:
$H(\omega) = \frac{1}{j\omega C_1 R_1 + 1}$ where $R_1 = 100\Omega$ and $C_1 = 100pF$

i. What is its cutoff frequency?
Hint: Recall that the cutoff frequency is the frequency at which the magnitude of the transfer function is $\frac{1}{\sqrt{2}}$.

ii. Sketch its phase and magnitude.

$|H(\omega)|$

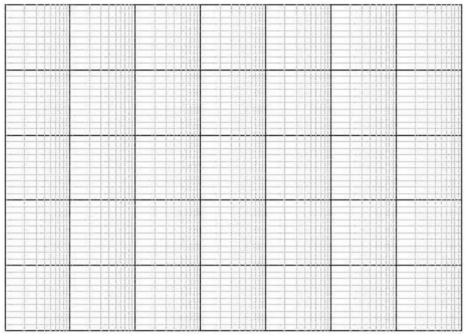
$\angle H(\omega)$

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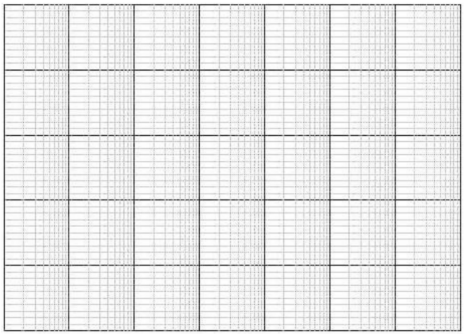
(b) Now consider the transfer function of a High-Pass Filter: $H(\omega) = \frac{j\omega C_2 R_2}{j\omega C_2 R_2 + 1}$ where $R_2 = 1k\Omega$ and $C_1 = 10nF$

i. What is its cutoff frequency?

ii. Sketch its phase and magnitude.



$|H(\omega)|$



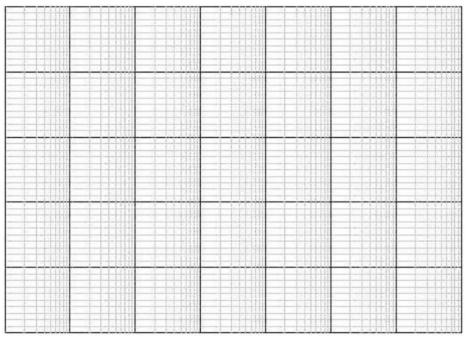
$\angle H(\omega)$

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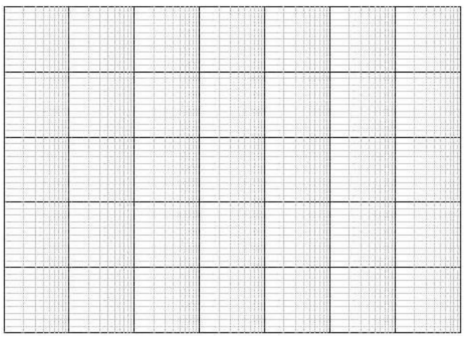
(c) What happens if we cascade these two filters together with a unity-gain buffer in between them? Consider the resulting transfer function: $H(\omega) = \frac{1}{j\omega C_1 R_1 + 1} \cdot \frac{j\omega C_2 R_2}{j\omega C_2 R_2 + 1}$

i. What are its cutoff frequencies?

ii. Sketch its phase and magnitude. *Hint: How can we combine the plots of the individual filters together?*



$|H(\omega)|$



$\angle H(\omega)$

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2. Bode Plots for Filters

Bode plots provide us with a simple and easy tool to plot these transfer functions by hand. **Always remember that Bode plots are an approximation**; if you want the precisely correct plots, you need to use numerical methods (like solving using MATLAB or IPython).

When we make Bode plots, we plot the frequency and magintude on a logarithmic scale and the angle in either degrees or radians. We use the log scale because it allows us to break up complex transfer functions into its constituent components.

When plotting the transfer function, the most important quantity to look at is its cutoff frequency ω_c. We will take a look the individual Bode plots for low and high pass filters, and then look at how the Bode plot for a bandpass filter is constructed.

(a) Consider a circuit with transfer function: $H(\omega) = \frac{1}{j\omega C_1 R_1 + 1} \cdot \frac{j\omega C_2 R_2}{j\omega C_2 R_2 + 1}$

i. What are the cutoff frequencies of this filter?

ii. Sketch its phase and magnitude. *Hint: How can we combine the plots of the individual filters together?*

Now lets take a look at two more concepts, **bandwidth** and **Q-factor**. The bandwidth of a resonance circuit is the difference between its upper and lower cutoff frequencies. It can also be expressed as *resonance frequency* divided by its *Q-factor*:

$$bandwidth = \Delta\omega = f_{high} - f_{low} = \frac{\omega_n}{Q} \tag{1}$$

The Q-factor, or quality factor, of a circuit is the ratio of power stored to power dissipated, but can also be more simply thought of as the measure of how good a circuit is, where a higher Q-factor is often more desirable. This can be expressed mathematically as:

$$Q = \frac{\omega_n}{\Delta\omega} \tag{2}$$

(b) Find the bandwidth of this circuit.

(c) What is its Q-factor? I.e.: $\frac{\omega_n}{\Delta\omega}$, where ω_n is the center frequency of the band.

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3. Filter Design

Suppose you have been hired to design a biomedical sensor that can detect and output recordings of Alpha brainwaves in the frequency range 8 Hz to 12 Hz. Unfortunately, our sensor is faulty: it is also picking up Gamma brainwaves in the frequency range 40 Hz to 100 Hz, interfering with our ability to get clean recordings of alpha brainwaves. We want to create a new design for our sensor that can remove this interference, giving us a clearer signal.

For the following, it's recommended that you graph the plots with a computer.

(a) What kind of filter could you use to remove this interference?

(b) Assume we only have access to resistors and capacitors. **Sketch the corresponding filter circuit and write out its transfer function.**

(c) $\omega_c = \frac{1}{RC}$ is the cutoff frequency which determines the frequency at which our filter starts attenuating the signal. **Should we maximize or minimize ω_c to remove as much of the interference as possible?**

(d) Say we can set our cutoff frequency to 10 Hz, 20 Hz, 32 Hz, 100 Hz, or 120 Hz. Which is the best cutoff frequency, and why?

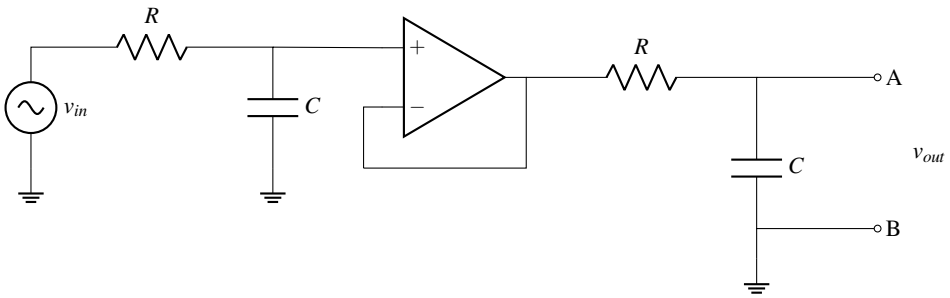
(e) We only have a 3.3 kΩ resistor in our workstation. What capacitor value should we use for our filter?

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(f) Use a computer to plot the magnitude and phase response of this filter.

(g) We can look at the magnitude phase plots of the previous part, realize that the performance our filter is rather poor. $|H(j\omega)|$ at 40 Hz is still around 0.5. This is because the Alpha and Delta brainwaves are very close together in frequency, and it takes an entire magnitude for the magnitude of the transfer function to drop by a factor of 10.

To combat this, we can cascade two of the low pass filters from the previous part with the same resistor and capacitor values, with a buffer in between as shown below:



What is the transfer function $H(j\omega)$, and see what the response looks like on a computer. In addition, what would happen if we cascaded a large number of filters together?

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(h) Now, let's say that our filter is successful at removing the interference we initially detected. We now notice that there is another source of interference–Delta brainwaves–in the frequency range 0.5Hz to 4Hz. **How could we modify our design to remove both sources of interference and get a clear recording of Alpha brainwaves?** Suppose you do not have access to an op-amp, but you do have access to an inductor.

(i) Where would you want to set the cutoff frequencies?

(j) How can we calculate the cutoff frequencies for our filter in terms of R, L, and C?

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(k) Let's pick values of R, L, and C to set our cutoff frequencies to those picked in part (i). Suppose you have a $500\,\Omega$ resistor. What values should you pick for your capacitior and inductor?

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(l) Plot the frequency response of the filter designed above using the resistor, capacitor and inductor values picked in the previous part. You should plot the frequency response on a computer.

(m) We again notice that while the filter is attenuating the necessary frequencies, it isn't doing that good of a job. Try plotting the responses if we had multiple filters cascaded with buffers similar to part (g).

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